This section provides knowledge and understanding of stars, Wien's displacement law, Stefan's law, Hubble's law and the Big Bang.

Learners have the opportunity to appreciate how scientific ideas of the Big Bang developed over time (HSW7) and how its validity is supported by experimental work carried out by the scientific community (HSW11).

5.5.1 Stars

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the terms planets, planetary satellites, comets, solar systems, galaxies and the universe	
(b)	formation of a star from interstellar dust and gas in terms of gravitational collapse, fusion of hydrogen into helium, radiation and gas pressure	Learners are not expected to know the details of fusion in terms of Einstein's mass-energy equation.
(c)	evolution of a low-mass star like our Sun into a red giant and white dwarf; planetary nebula	
(d)	characteristics of a white dwarf; electron degeneracy pressure; Chandrasekhar limit	
(e)	evolution of a massive star into a red super giant and then either a neutron star or black hole; supernova	
(f)	characteristics of a neutron star and a black hole	
(g)	Hertzsprung–Russell (HR) diagram as luminosity–temperature plot; main sequence; red giants; super red giants; white dwarfs.	